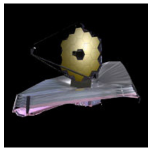

5765 - Two for the price of one: complete spectroscopy from 0.7 to 23 micron of the benchmark brown dwarfs Eps Ind BA and BB

Cycle: 3, Proposal Category: GO

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
Observation Folder				
	1	Eps-INDI-MRS	MIRI Medium Resolution Spectroscopy	(1) -eps-Ind-B
	2	Eps-INDI-NIRSpec	NIRSpec IFU Spectroscopy	(1) -eps-Ind-B

ABSTRACT

The binary brown dwarf Eps Ind BA/BB is a key benchmark system with well measured mass, age and metallicity, and serves as a unique laboratory for comparative studies of brown dwarf atmospheres. We will use spectroscopic observations of Eps Ind BA/BB to carry out several crucial tests of brown dwarf formation, evolution and atmospheric chemistry, such as the following: a stringent comparison of mass, bolometric luminosity, and age against models; measurement of the 15N/14N ratio as a tracer of formation; testing the ability of retrievals to infer accurate C/O ratios; a comparison of brown dwarf cloud properties, in a region where silicate clouds in the atmosphere are expected to disperse. This work will underpin broader efforts to understand brown dwarf and exoplanet atmospheres, and the tail end of the IMF.

Eps Ind BA/BB are the 2nd and 3rd closest T dwarfs to earch, allowing excellent signal-to noise spectroscopic measurements. The pair has a well-constrained orbit giving extremely precise mass measurements (<0.5% accuracy), and a favorable semi-major axis allowing the pair to be spatially resolved with JWST. They are co-moving with a main-sequence star, providing an independent measure of the system age and metallicity. That star itself hosts a wide-orbit super-Jupiter planet - potentially allowing for the comparative studies of three substellar companions with the same age, metallicity and formation environment.

We will collect a complete spectrum of the brown dwarf binary from 0.7um to ~23um with NIRSpec and MIRI, spatially resolved to ~12um. These observations will provide a rich legacy JWST dataset.

OBSERVING DESCRIPTION

We will collect spectroscopic observations of Eps Ind BA/BB using the NIRSpec Integral Field Unit (IFU) and the MIRI Medium Resolution Spectrometer. We will center the brown dwarf in the field of view.

The NIRSpec IFU observation will use the G140H, G235H and the G395H grisms, with an exposure time of 900s for each grism. We will use four dither positions to improve sensitivity and reduce detector effects.

The MIRI MRS observation will be performed in all four channels with all three dispersers, and we will observe for 777s for each disperser (divided over four dither positions). In addition simultaneous imaging will be performed with the MIRI imager to provide astrometry that will enable accurate reconstruction of the dithered MRS observations. Similarly to the NIRSpec IFU observation, the dither positions will improve the sensitivity and reduce detector effects.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	-eps-Ind-B	RA: 22 04 10.5952 (331.0441467d) Dec: -56 46 58.24 (-56.78284d) Equinox: J2000	Proper Motion RA: 3981.9769999999994 mas/yr Proper Motion Dec: -2466.8319999818777 mas/yr Parallax: 0.270658" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Brown dwarfs]

Observation

Proposal 5765, Observation 1: Eps-INDI-MRS

Mon Jul 28 22:00:18 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	-eps-Ind-B	RA: 22 04 10.5952 (331.0441467d) Dec: -56 46 58.24 (-56.78284d) Equinox: J2000	Proper Motion RA: 3981.9769999999994 mas/yr Proper Motion Dec: -2466.8319999818777 mas/yr Parallax: 0.270658" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Brown dwarfs]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	FND	FAST	10	1	1	27.75	173206

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F1000W	FASTR1	10	5	1	Dither 1	4	20	599.409	
1	SHORT(A)	MRSLONG		FASTR1	75	1	1	Dither 1	4	4	832.512	
1	SHORT(A)	MRSSHORT		FASTR1	75	1	1	Dither 1	4	4	832.512	
2		IMAGER	F770W	FASTR1	10	5	1	Dither 1	4	20	599.409	
2	MEDIUM(B)	MRSLONG		FASTR1	75	1	1	Dither 1	4	4	832.512	
2	MEDIUM(B)	MRSSHORT		FASTR1	75	1	1	Dither 1	4	4	832.512	
3		IMAGER	F560W	FASTR1	10	5	1	Dither 1	4	20	599.409	
3	LONG(C)	MRSLONG		FASTR1	75	1	1	Dither 1	4	4	832.512	
3	LONG(C)	MRSSHORT		FASTR1	75	1	1	Dither 1	4	4	832.512	

Special Requirements	
	After Date 01-SEP-2025:00:00:00

Observation

Proposal 5765, Observation 2: Eps-INDI-NIRSpec

Mon Jul 28 22:00:18 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	-eps-Ind-B	RA: 22 04 10.5952 (331.0441467d) Dec: -56 46 58.24 (-56.78284d) Equinox: J2000	Proper Motion RA: 3981.9769999999994 mas/yr Proper Motion Dec: -2466.8319999818777 mas/yr Parallax: 0.270658" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Brown dwarfs]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395H/F290LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	
2	G235H/F170LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	
3	G140H/F100LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	

Special Requirements

After Date 01-SEP-2025:00:00:00